

Ayush Kumar Dubey

 [dubeyayush-exe](#) |  [Ayush Dubey](#) |  [Portfolio](#) |  ayushdubey421@gmail.com

EDUCATION

Bachelor of Technology: Artificial Intelligence and Machine Learning 2022 - 2026
Chandigarh University
Relevant Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Data Structures & Algorithms, DBMS

TECHNICAL SKILLS

ML/AI Computer Vision (ResNet, YOLO, Detection/Segmentation), NLP, Time Series Forecasting, Model Optimization
Languages Python, C++, Java, SQL
Frameworks PyTorch, TensorFlow, OpenCV, Scikit-learn, FastAPI, LangChain, Ollama
MLOps MLflow, Docker, Prometheus, Grafana, Redis, CI/CD (GitHub Actions)
Cloud/Tools AWS, Git, Linux, PostgreSQL, MySQL

PROFESSIONAL EXPERIENCE

AI/ML Intern, Defense Research and Development Organization (DRDO) 2022 - 2026

- Engineered 3D GRU-based scene reconstruction pipeline using PyTorch, improving accuracy from 48.1% to 90.8% (42.7% gain) for defense-grade vision systems deployed in surveillance applications
- Collaborated with cross-functional teams (signal processing experts, physicists) to optimize model latency to 180ms/frame while maintaining 85%+ accuracy on real-world defense datasets

Machine Learning Intern, Patanjali Ayurveda Pvt. Ltd. 2022 - 2026

- Deployed ResNet-18 CNN for biometric authentication, reducing false rejection rate by 12.4% (from 18.2% to 5.8%), improving system reliability for 3,000+ daily active users across 15+ facilities
- Built SARIMAX forecasting model with exogenous features (promotions, seasonality) reducing MAPE by 15.6% on 48 months of sales data, directly informing \$2M+ inventory planning decisions

TECHNICAL PROJECTS

LocalMind: Hybrid AI Assistant with MLOps Pipeline

- Built production-grade CLI assistant using Ollama (Llama 3.2) for offline inference and Claude API for complex queries, achieving 73% local routing rate and \$2.34/month cost savings vs API-only approach
- Implemented MLOps infrastructure with MLflow (experiment tracking), Prometheus + Grafana (monitoring 15+ metrics including latency, accuracy, cost per query), demonstrating 99.2% uptime over 30 days
- Architected intelligent routing system using lightweight classifier (XGBoost) deciding local vs cloud execution based on query complexity, reducing average response latency to 340ms

PulsePay: Health ML Platform with Subscription System

- Developing cardiovascular risk prediction system using XGBoost on 10K+ patient records, achieving 84% AUC with SHAP explainability for clinical interpretability
- Building automated ML pipeline with model versioning (MLflow), A/B testing framework, and drift detection (Evidently), reducing model deployment time from manual 2-day process to automated 15-minute CI/CD
- Integrated Stripe subscription API with webhook handling for payment processing, implementing feature-gating logic and usage-based billing for 3-tier SaaS model

RESEARCH & PUBLICATIONS

Heart Failure Prediction using Artificial Neural Networks
Published at ICETET-SIP 25 (Paper ID: 753) — Developed ANN classifier achieving 87% accuracy and 0.82 AUC on UCI Heart Failure dataset for early clinical intervention, with feature importance analysis using SHAP values

LEADERSHIP & ACTIVITIES

Joint Secretary, Ashura Club: Organized 12 technical workshops and hackathons in my college, leading a team of 24 members and engaging 200+ students in ML/AI learning initiatives